MATH 152
Mrs. Bonny Tighe

QUIZ 9

25 points
11.10 – 11.11

NAME Solutions

TA_____ Wed 11/20/2019

This is a take-home quiz. You may not ask Janita, Jessie or Mrs. Tighe for help on this quiz. You must turn the quiz in before you take the first half of your Exam on Monday morning in discussion.

4pts

1. Evaluate the limit using series $\lim_{x\to 0} \dfrac{x^2 - \sin(x^2)}{e^{x^3} - (1+x^3)}$

$$\sin(x^2) = \sum_{n=0}^{\infty} \frac{(-1)^n (x^2)^{2n+1}}{(2n+1)!} \qquad 4n+2$$

$$e^{x^3} = \sum_{n=0}^{\infty} \frac{(x^3)^n}{n!} = x^{3n}$$

$$\lim_{x\to 0} \frac{x^2 - \left[x^2 - \frac{x^6}{3!} + \frac{x^{10}}{5!} - \frac{x^{14}}{7!} \cdots\right]}{1 + x^3 + \frac{x^6}{2!} + \frac{x^9}{3!} + \cdots - [1+x^3]}$$

$$\lim_{x\to 0} \left(\frac{\frac{x^6}{3!} - \frac{x^{10}}{5!} + \frac{x^{14}}{7!} \cdots}{\frac{x^6}{2!} + \frac{x^9}{5!} + \frac{x^{12}}{7!} \cdots} \right) = \lim_{x\to 0} \frac{\cancel{x^6}\left[\frac{1}{6} - \frac{x^4}{5!} + \frac{x^8}{7!} + \cdots\right]}{\cancel{x^6}\left[\frac{1}{2} + \frac{x^3}{5!} + \frac{x^6}{7!} + \cdots\right]}$$

$$\frac{\frac{1}{6}}{\frac{1}{2}} = \frac{1}{6}\left(\frac{2}{1}\right) = \frac{1}{3}$$

2. Use the Binomial Theorem for the expansion of the function as a power series.

4pts

$$f(x) = \frac{1}{(1+3x^2)^3} = (1+3x^2)^{-3} \qquad \sum_{n=0}^{\infty} \square \frac{(3x^2)^n}{n!}$$

$$\square = 1 - 3 - 3(-4) - 3(-4)(-5) - 3(-4)(-5)(-6)$$

$$= \underset{n=0}{1} - \underset{n=1}{\frac{3!}{2}} + \underset{n=2}{\frac{4!}{2}} - \underset{n=3}{\frac{5!}{2}} + \frac{6!}{2} \cdots \qquad (-1)^n \frac{(n+2)!}{2}$$

$$\sum_{n=0}^{\infty} \frac{(-1)^n (n+2)(n+1)n!\, 3^n x^{2n}}{2\, n!} = \boxed{\sum_{n=0}^{\infty} \frac{(-1)^n (n+2)(n+1) 3^n x^{2n}}{2}}$$

3. Find the MacLaurin series using a known series

2pts $f(x) = 9\cos(3x^3)$

$$\cos x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!}$$

$$3^2 \sum_{n=0}^{\infty} \frac{(-1)^n (3x^3)^{2n}}{(2n)!} = 3^2 \sum_{n=0}^{\infty} \frac{(-1)^n 3^{2n} x^{6n}}{(2n)!}$$

$$\boxed{\sum_{n=0}^{\infty} \frac{(-1)^n 3^{2n+2} x^{6n}}{(2n)!}}$$

4. Find the Taylor series for $f(x) = \dfrac{1}{\sqrt[3]{5x+2}}$ centered at $a = 5$

$$\sum_{n=0}^{\infty} \frac{f^{(n)}(5)(x-5)^n}{n!}$$

4 pts

$f(x) = (5x+2)^{-1/3}$

$f'(x) = -\frac{1}{3}(5x+2)^{-4/3}(5)$

$f''(x) = \frac{4}{3^2}(5x+2)^{-7/3}(5^2)$

$f'''(x) = \frac{-4\cdot 7}{3^3}(5x+2)^{-10/3}(5^3)$

$f''''(x) = \frac{1\cdot 4\cdot 7\cdot 10}{3^4}(5x+2)^{-13/3}(5^4)$

$f(5) = 27^{-1/3} = \frac{1}{3}$

$f'(5) = -\frac{1}{3}\cdot\frac{1}{3^4}(5)$

$f''(x) = \frac{1\cdot 4}{3^2}\cdot\frac{1}{3^7}(5^2)$

$f'''(x) = \frac{-1\cdot 4\cdot 7}{3^3}\cdot\frac{1}{3^{10}}(5^3)$

$f''''(x) = \frac{1\cdot 4\cdot 7\cdot 10}{3^4}\cdot\frac{1}{3^{13}}(5^4)$

$$\underset{n=0}{\frac{1}{3}} - \underset{n=1}{\frac{1}{3^5}} + \underset{n=2}{\frac{1\cdot 4}{3^9}} - \underset{n=3}{\frac{1\cdot 4\cdot 7}{3^{13}}} + \underset{n=4}{\frac{1\cdot 4\cdot 7\cdot 10}{3^{17}}}(5^4)$$

$$\frac{(-1)^n(1\cdot 4\cdot 7\cdots(3n-2))}{3^{4n+1}}$$

$$\frac{1}{3} + \sum_{n=1}^{\infty} \frac{(-1)^n(1\cdot 4\cdot 7\cdots(3n-2))5^n}{3^{4n+1}\, n!}(x-5)^n$$

(1pt)

5. Find the sum of the power series a) $\sum_{k=0}^{\infty} \frac{(-1)^k 3^k}{(2k+1)}$

$3^k = (\sqrt{3})^{2k} = \frac{(\sqrt{3})^{2k+1}}{\sqrt{3}}$

$\sum_{k=0}^{\infty} \frac{(-1)^k (\sqrt{3})^{2k+1}}{\sqrt{3}\,(2k+1)} = \frac{1}{\sqrt{3}} \tan^{-1}(\sqrt{3}) = \boxed{\frac{1}{\sqrt{3}}\left(\frac{\pi}{3}\right)}$

(1pt) b) $\sum_{k=0}^{\infty} \frac{(-1)^k \pi^{2k}}{36^k (2k)!}$

6^{2k}

$\sum_{k=0}^{\infty} \frac{(-1)^k (\pi/6)^{2k}}{(2k)!} = \cos \pi/6 = \sqrt{3}/2$

6. Find the Taylor polynomial $T_4(x)$ for the function $f(x) = \sin x$ at $a = \pi/6$ and use it to estimate $\sin(29\pi/180)$

(3pts)

$\sum_{n=0}^{\infty} \frac{f^{(n)}(\pi/6)(x-\pi/6)}{n!}$

$f(x) = \sin x$	$f(\pi/6) = 1/2$
$f'(x) = \cos x$	$f'(\pi/6) = \sqrt{3}/2$
$f''(x) = -\sin x$	$f''(\pi/6) = -1/2$
$f'''(x) = -\cos x$	$f'''(\pi/6) = -\sqrt{3}/2$
$f''''(x) = \sin x$	$f''''(\pi/6) = 1/2$

$\frac{29\pi}{180} - \frac{\pi}{6} = \frac{-\pi}{180}$

$T_4(x) = \frac{1}{2} + \frac{\sqrt{3}}{2}(x-\pi/6) - \frac{1}{2}\frac{(x-\pi/6)^2}{2} - \frac{\sqrt{3}}{2}\frac{(x-\pi/6)^3}{3!} + \frac{1}{2}\frac{(x-\pi/6)^4}{4!}$

$\sin(29\pi/180) = \frac{1}{2} + \frac{\sqrt{3}}{2}\left(\frac{-\pi}{180}\right) - \frac{1}{2}\frac{\left(\frac{-\pi}{180}\right)^2}{2} - \frac{\sqrt{3}}{2}\frac{(-\pi/180)^3}{3!} + \frac{1}{2}\frac{(-\pi/180)^4}{4!}$

7. Use the **TAYLOR'S INEQUALITY** If $\left|f^{(n+1)}(x)\right| \le M$ for $|x-a| \le d$, then the remainder $R_n(x) \le \frac{M}{(n+1)!}|x-a|^{n+1}$ for $|x-a| \le d$

$$\sum_{n=0}^{\infty} \frac{f^{(n)}(3)(x-3)^n}{n!}$$

3 pts Approximate the function $f(x) = \ln(1+3x)$ by a Taylor polynomial of degree 3 at $a = 3$

$f(x) = \ln(1+3x)$

$f'(x) = (1+3x)^{-1}(3)$

$f''(x) = -(1+3x)^{-2}(3^2)$

$f'''(x) = 2(1+3x)^{-3}(3^3)$

$f''''(x) = -3!(1+3x)^{-4}(3^4)$

$f(3) = \ln(10)$

$f'(3) = \frac{1}{10} \cdot 3$

$f''(x) = \frac{-1}{10^2}(3^2)$

$f'''(x) = \frac{2!}{10^3}(3^3)$

$$\ln(10) + \frac{3}{10}(x-3) - \frac{3^2}{10^2}\frac{(x-3)^2}{2} + \frac{2(3^3)(x-3)^3}{10^3 3!} = T_3(x)$$

3 pts How accurate is this approximation when $2.5 < x < 3.5$

$|x-3| < .5$

$d = .5$

$-.5 < x-3 < .5$

$$\left|f''''(x)\right| = \left|\frac{-6(81)}{(1+3x)^4}\right|$$

$M = \frac{486}{(1+3x)^4}$ is largest when $x = 2\frac{1}{2} = 5/2$

$$M = \frac{486}{(1+3(5/2))^4} = \frac{486}{(17/2)^4}$$

$$R_3(x) \le \frac{486}{(17/2)^4}\frac{|.5|^4}{4!} \text{ or } \frac{486(2^4)}{(17)^4 4!}\left(\frac{1}{2}\right)^4$$